NATIONAL UNIVERSITY OF SINGAPORE

SEMESTER I, 2026/2027

MA1522 Linear Algebra for Computing **Graded Quiz 1**

1. Which of the following are linear equations?

(a) $5x + \left(\sin\dfrac{5\pi}{12}\right) y = e^2 z + 2\cos\dfrac{\pi}{12}$.

(b) $5x + \left(\sin\dfrac{5\pi}{12}\right) y = e^{2z} + 2\cos\dfrac{\pi}{12}$.

(c) $5x + \sin\dfrac{5\pi y}{12} = e^2 z + 2\cos\dfrac{\pi}{12}$.

(d) $5x + \left(\sin\dfrac{5\pi}{12}\right) y = e^2 z$.

Solution. A linear equation in variables x, y, z has the form

$$ax + by + cz = d$$

where a, b, c are (real) constants. So only (a) and (d) are linear equations.

2. Let $\boldsymbol{A}$ be an augmented matrix, and let R_i and R_j denote the ith and j^{th} row of $\boldsymbol{A}$ respectively, where $i \neq j$.

Which of the following row operations on $\boldsymbol{A}$ produce a matrix that is row equivalent to $\boldsymbol{A}$?

(a) Adding cR_j to R_i for any real number c.

(b) Multiplying R_i by 0.

(c) Interchanging R_i and R_j.

(d) Replacing R_i by $cR_i + dR_j$ for any real numbers c and d with $c \neq 0$.

Solution. (a) True. This is the elementary row operation $R_i + cR_j$.

(b) False. If a row is multiplied by 0, the restriction provided by R_i may be lost.

(c) True. This is the elementary row operation $R_i \leftrightarrow R_j$.

(d) True. This is the composite of elementary row operations cR_i followed by $R_i + dR_j$.

3. For a linear system of m linear equations in n unknowns to be consistent, which of the following statements are true?

(a) It is necessary that $m \geq n$.

(b) It is sufficient that $m \geq n$.

(c) It is necessary that $m \leq n$.

(d) It is sufficient that $m \leq n$.

Solution. If a linear system has an inconsistent equation $0x_1 + \cdots + 0x_n = b$ ($b \neq 0$), then the system is inconsistent. Hence, for any values of m and n, a linear system of m equations in n unknowns may be consistent or inconsistent.

Hence, none of the above is true.

4. For a linear system of m linear equations in n unknowns to have a unique solution, which of the following statements are true?

(a) It is necessary that $m = n$.

(b) It is necessary that $m \geq n$.

(c) It is sufficient that $m \geq n$.

(d) It is sufficient that the system is consistent and $m \geq n$.

Solution. Let $(\boldsymbol{R} \mid \boldsymbol{b})$ be a row-echelon form of the associated augmented matrix.

If the system has a unique solution, then $\boldsymbol{b}$ is non-pivot, and all columns of $\boldsymbol{R}$ are pivot. So it is necessary that

$$\begin{aligned} m &= \text{number of rows of } \boldsymbol{R} \\ &\geq \text{number of nonzero rows of } \boldsymbol{R} \\ &= \text{number of pivot columns of } \boldsymbol{R} = n. \end{aligned}$$

Conversely, if a system has infinitely many solutions, zero equations can be added to the system so that the new system have $m \geq n$ and still have the same solution set (and therefore is consistent). Hence, $m \geq n$, even with the additional condition that the system is consistent, is not sufficient for the system to have a unique solution.

Therefore, only (b) is true.

5. Which of the following (augmented) matrices are in row-echelon form?

(a) $\left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$.

(b) $\left(\begin{array}{ccc|c} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right)$.

(c) $\left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$.

(d) $\left(\begin{array}{ccc|c} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right)$.

Solution. (a) and (b) are in row-echelon form, but (c) and (d) are not.

6. Which of the following matrices are in reduced row-echelon form?

(a) $\left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$.

(b) $\left(\begin{array}{cccc} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right)$.

(c) $\left(\begin{array}{cccc} 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right)$.

(d) $\left(\begin{array}{cccc} -1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right)$.

Solution. (a) and (b) are in reduced row-echelon form, but (c) and (d) are not.

7. Find the number of solutions to the linear system whose associated augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 2 & -4 & 5 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

Solution. The matrix $\left(\begin{array}{ccc|c} 1 & 2 & -4 & 5 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right)$ is in row-echelon form such that the rightmost column is non-pivot and the 3rd column is non-pivot. So the corresponding linear system has infinitely many solutions.

8. Find the number of solutions to the linear system whose associated augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 2 & -4 & 5 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 1 \end{array}\right).$$

Solution. The matrix $\left(\begin{array}{ccc|c} 1 & 2 & -4 & 5 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 1 \end{array}\right)$ is in row-echelon form such that the rightmost column is pivot. So the corresponding linear system is inconsistent.

9. Find the number of solutions to the linear system whose associated augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 2 & -4 & 5 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 1 & 0 \end{array}\right).$$

Solution. The matrix $\left(\begin{array}{ccc|c} 1 & 2 & -4 & 5 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 1 & 0 \end{array}\right)$ is in row-echelon form such that the rightmost column is non-pivot and all other columns are pivot. So the corresponding linear system has exactly one solution.

10. A system of 3 linear equations in variables x_1, x_2, x_3, x_4 has the following general solutions:

$$\begin{cases} x_1 = 5 + t \\ x_2 = 2 - t \\ x_3 = t \\ x_4 = 4 \end{cases}, \qquad t \in \mathbb{R}.$$

What is the reduced row-echelon form of the augmented matrix of the system?

Solution. Substitute the third equation into the first two, we get

$$\begin{cases} x_1 = 5 + x_3 \\ x_2 = 2 - x_3 \end{cases} \Rightarrow \begin{cases} x_1 - x_3 = 5 \\ x_2 + x_3 = 2 \end{cases}.$$

Adding the fourth equation, we get

$$\begin{cases} x_1 & - & x_3 & & = 5 \\ & x_2 & + & x_3 & = 2 \\ & & & x_4 & = 4 \end{cases}.$$

The augmented matrix of this system is

$$\left(\begin{array}{cccc|c} 1 & 0 & -1 & 0 & 5 \\ 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right)$$

which is already in reduced row-echelon form.